Slicing Is All You Need

Towards a Universal One-Sided Algorithm for Distributed Matrix Multiplication

Benjamin Brock

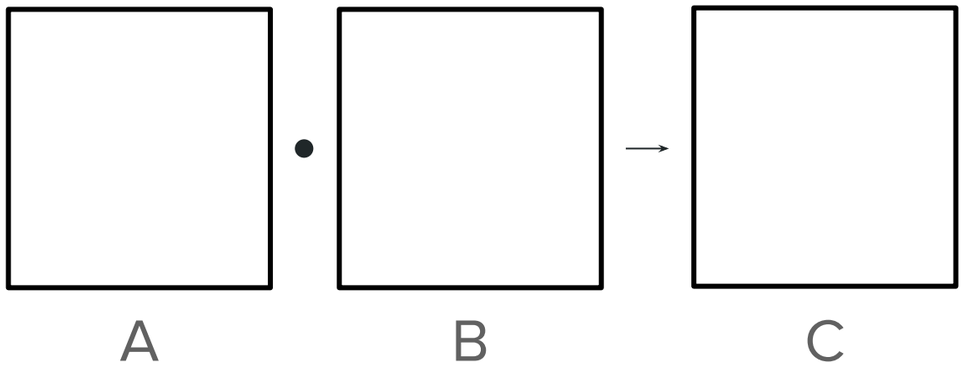
November 16, 2025

Distributed Matrix Multiplication

When a matrix is **too big** to fit on a **single node/GPU**

When computation **takes too long**

⇒ **Partition** matrix across multiple GPUs



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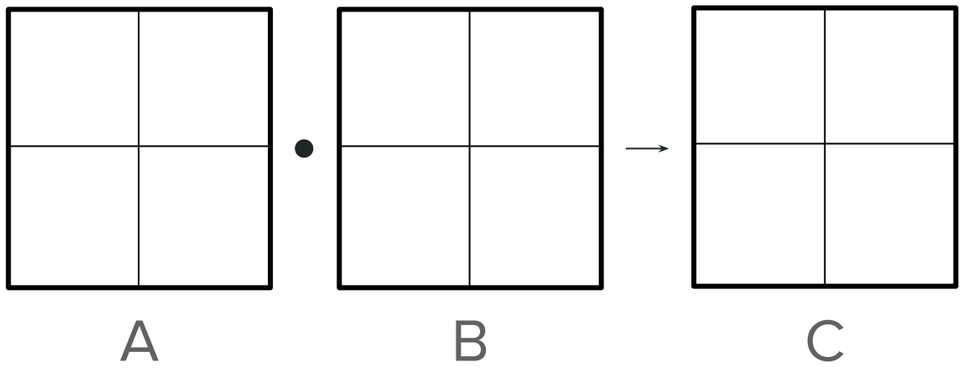

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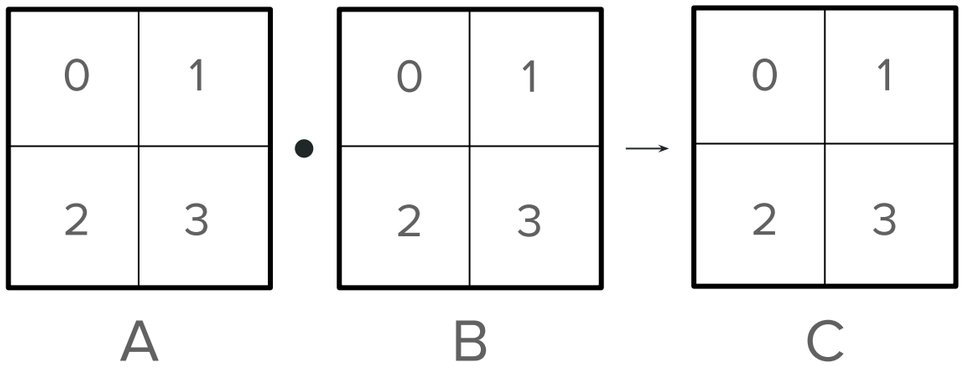

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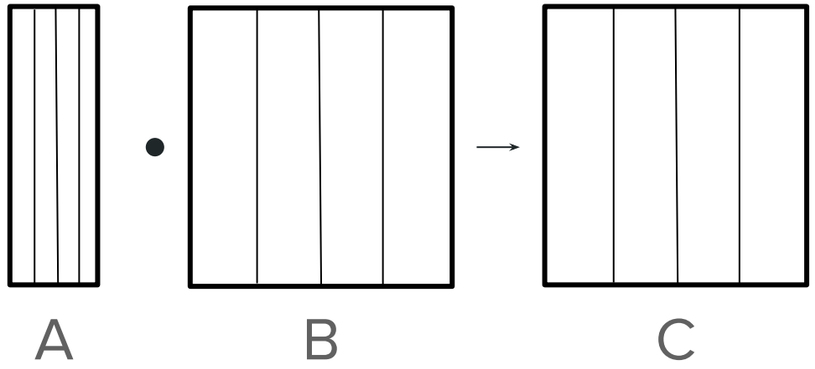
Distributed Matrix Multiplication

Different **problem sizes** require **different partitionings**

This is because **partitioning** determines **communication behavior**



A is small ⇒ pick a partitioning that only communicates A.

(1D column partitioning)

Distributed Matrix Multiplication

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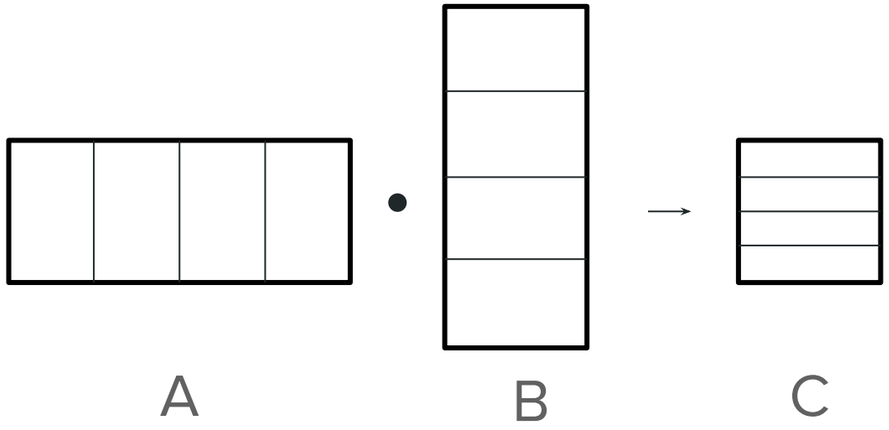
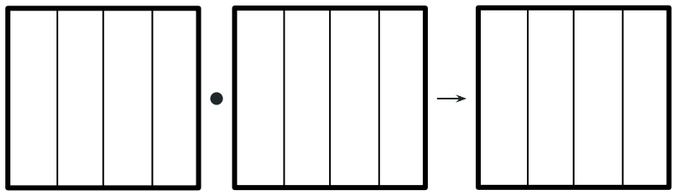


C is small ⇒ pick a partitioning that only communicates C.

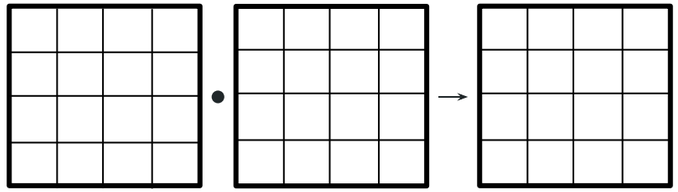
(Outer product partitioning)

Flavors of Distributed Matrix Multiplication

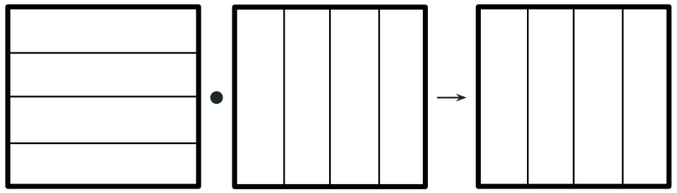


Column Block

Row Block



2D Distribution



Inner Product

Outer Product

+ cyclic distributions, replication

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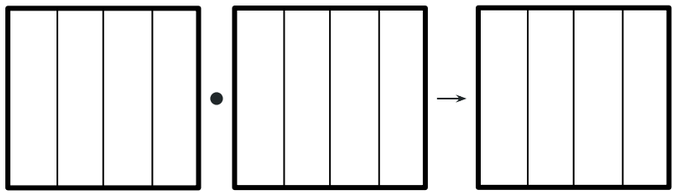
Flavors of Distributed Matrix Multiplication



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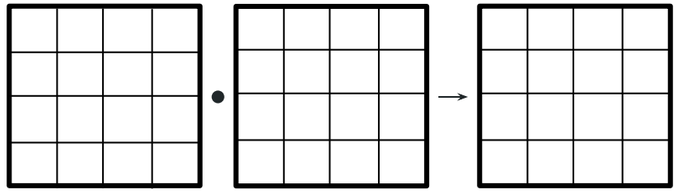

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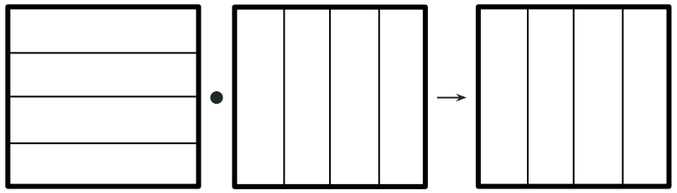

Inner Product

Outer Product

+ cyclic distributions, replication

Different distributions suitable for **different problems.**

Must **implement distributed algorithm** for **each variant**.

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SPMD Systems: Describe Partitionings

AI frameworks have developed "**SPMD**" **systems**, which allow users to set partitionings

SPMD systems then use **dispatching logic** to invoke an appropriate implementation

If no implementation is available, must **reshard** matrices

```
from torch.distributed.tensor import DeviceMesh,
Replicate, Shard, randn

# Create `a` and `b` with row block partitioning
a = randn((m,k),
          DeviceMesh('xpu', (world_size,)),
          [Shard(0)])
b = randn((k,n),
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# Perform distributed MatMul
c = torch.matmul(a, b)
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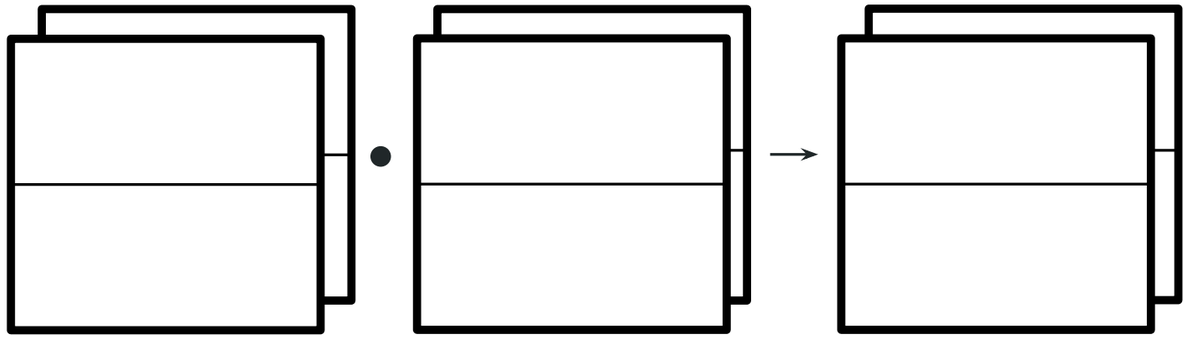

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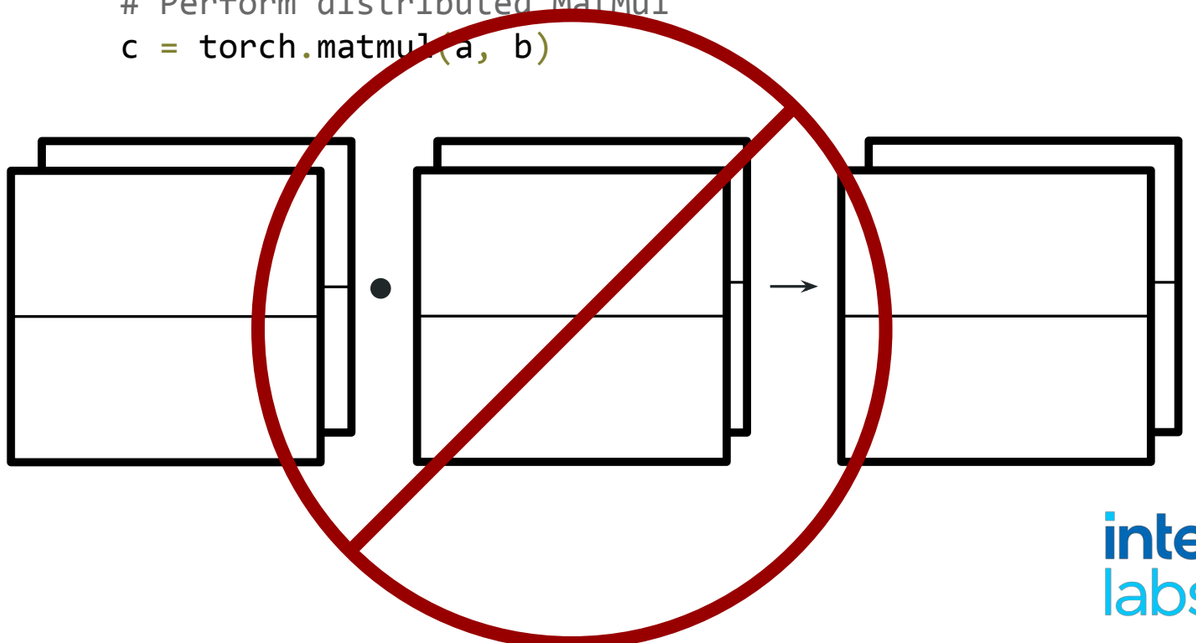
Not supported!



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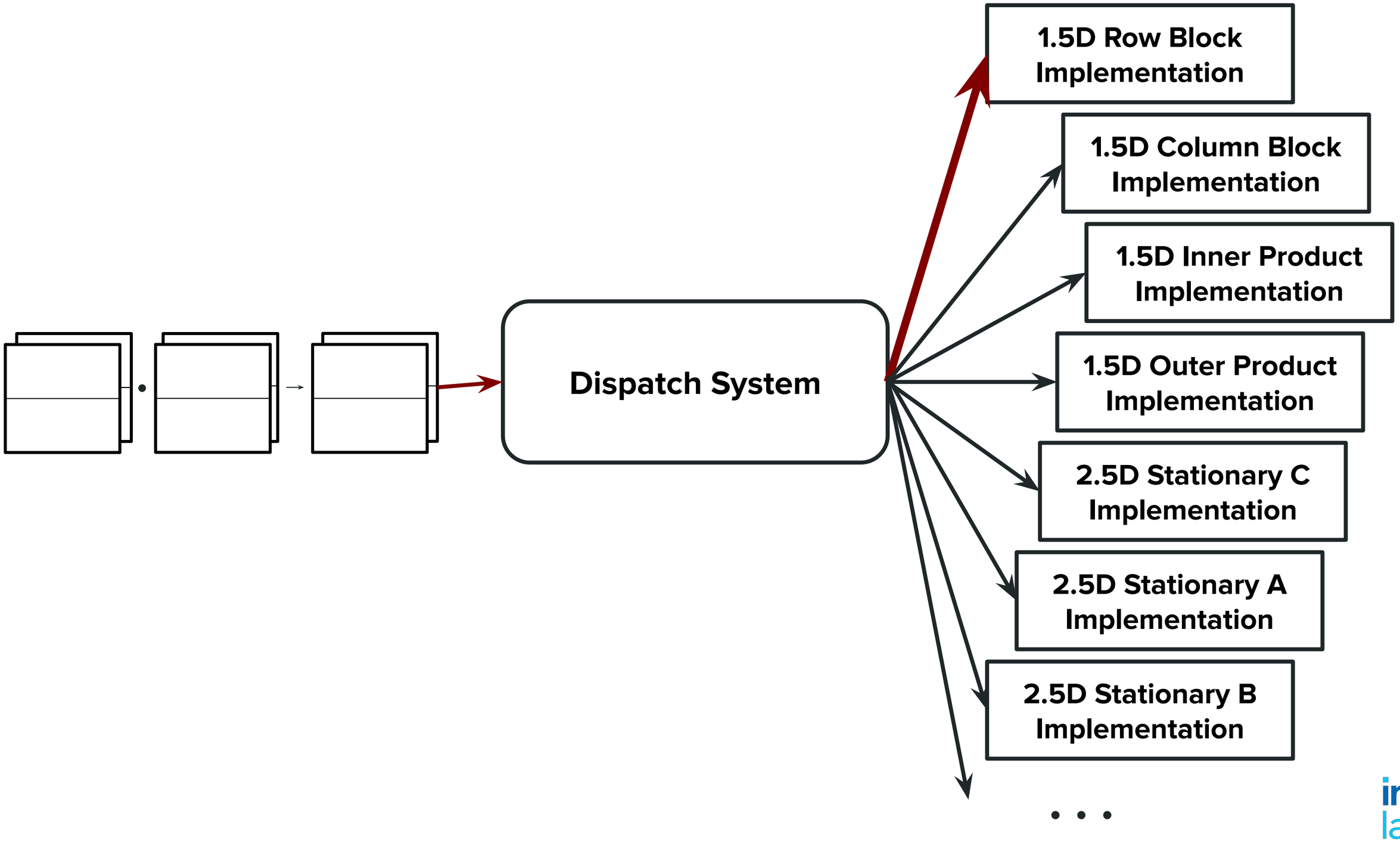
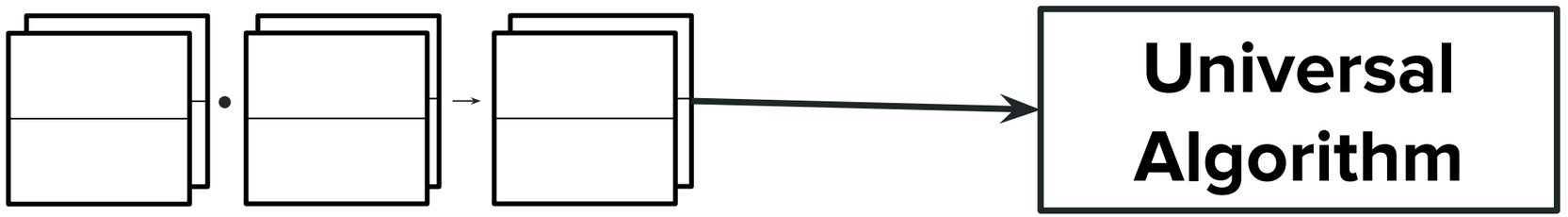

Dispatch System
1.5D Row Block Implementation
1.5D Column Block Implementation
1.5D Inner Product Implementation
1.5D Outer Product Implementation
2.5D Stationary C Implementation
2.5D Stationary A Implementation
2.5D Stationary B Implementation
. . .



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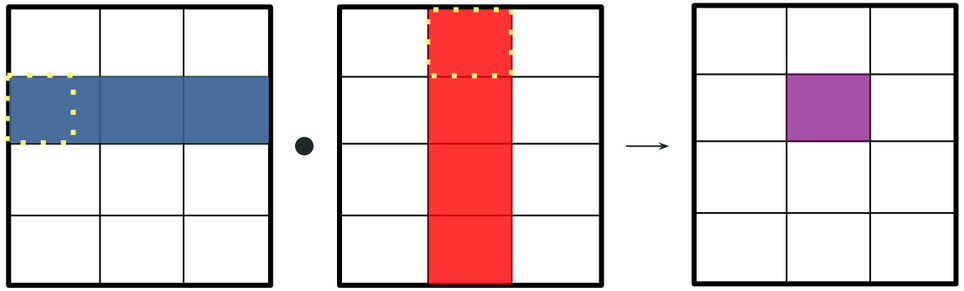
Slicing-Based Universal Algorithm

- Instead of **naively fetching** and **multiplying** tiles, **construct computation graph** of tiles to be multiplied.



Rank 0

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C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,0:32] * B.tile(0,0)[0:32,0:43]
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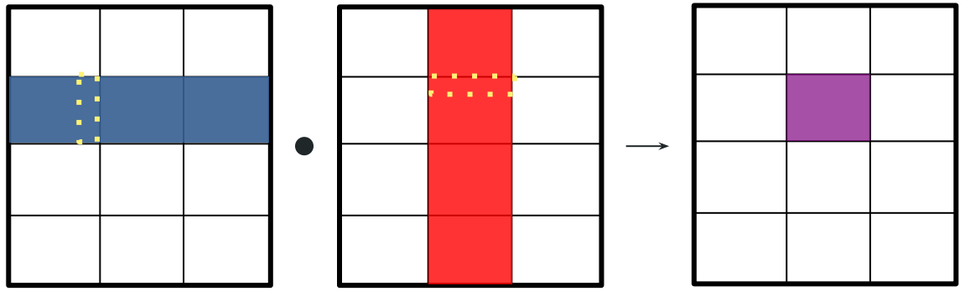

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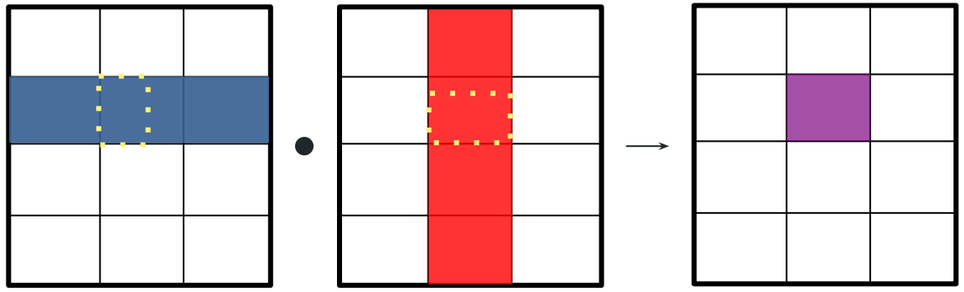

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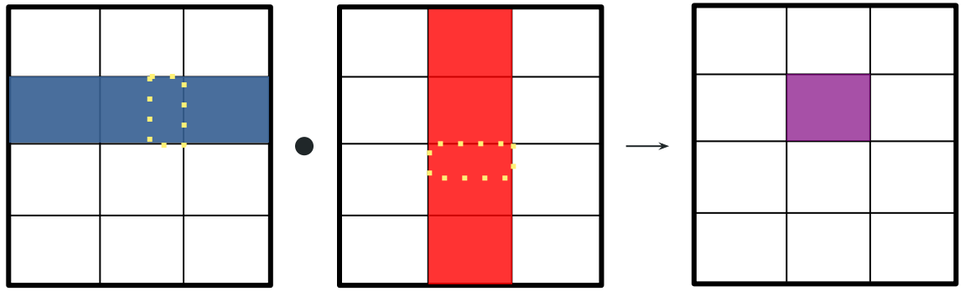

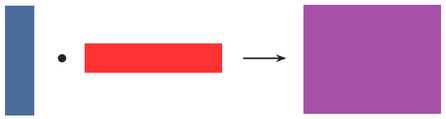

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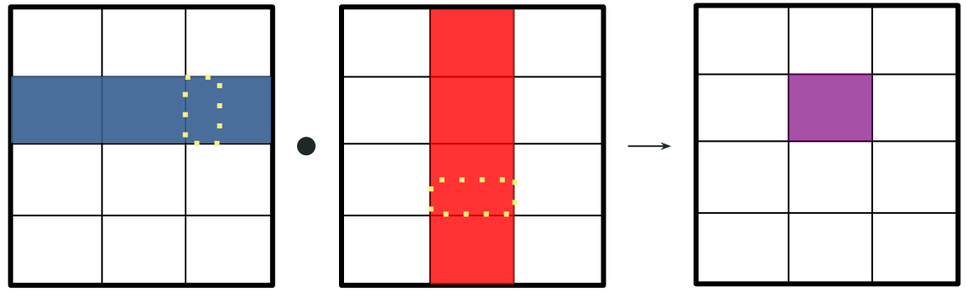

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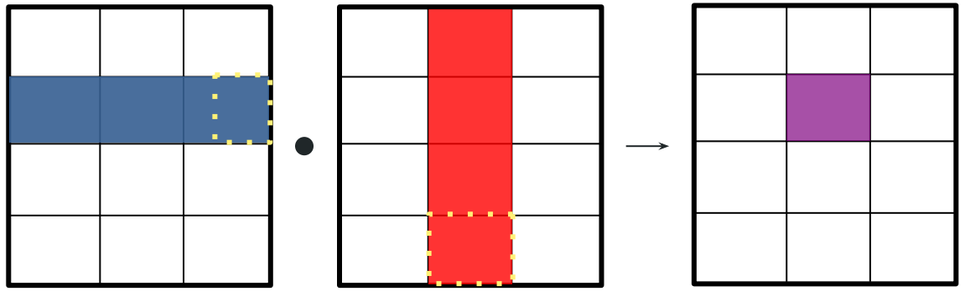

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Universal RDMA Algorithm: Optimizations

- Instead of **naively fetching** and **multiplying** tiles, **construct computation graph** of tiles to be multiplied.

Pseudocode: Produce IR

```
# Each process produces a list of operations to perform.
for idx,block in 'blocks of C I own':
    # Compute the bounds of my tile
    c_bounds = c.tile_bounds(idx)

    # Find tiles of A that overlap.
    a_tiles = a.overlapping_tiles(c_bounds[0],:)

    for a_idx in a_tiles:
        # Find tiles of B that overlap with our output bounds (of C)
        # *And* the current A tile's bounds.
        a_tile_bounds = a.tile_bounds(a_idx)
        b_tiles = b.overlapping_tiles(a_tile_bounds[1], c_bounds[1])

        for b_idx in b_tiles:
            # Emit instruction that multiplies tiles of
            # A, B, and C, including the appropriate slicing
            # of these tiles.
```

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Universal RDMA Algorithm: Optimizations

- Instead of **naively fetching** and **multiplying** tiles, **construct computation graph** of tiles to be multiplied.

- This graph can be **reordered** and **modified** to implement **computation/communication overlap**, prevent fetching tiles multiple times.

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C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,0:32] * B.tile(0,0)[0:32,0:43]
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```

This graph can be used to modified to implement computation/communication overlap, prevent fetching tiles multiple times.

```
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(0,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:32] * b_tile[0:32,0:43]
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,32:43] * b_tile[0:11,0:43]
a_tile = A.get_tile(0,1)
b_tile = B.tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:21] * b_tile[11:32,0:43]
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```

Universal RDMA Algorithm: Optimizations

Redundant Communications

- Instead of naively fetching and multiplying tiles, **construct computation graph** of tiles to be multiplied.
- This graph can be traversed and modified to implement **computation/communication overlap**, prevent fetching tiles multiple times.

Rank 0

```
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,0:32] * B.tile(0,0)[0:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,32:43] * B.tile(1,0)[0:11,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,0:21] * B.tile(1,0)[11:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,21:43] * B.tile(2,0)[0:22,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,0:10] * B.tile(2,0)[22:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,10:42] * B.tile(3,0)[0:32,0:43]
```

```
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(0,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:32] * b_tile[0:32,0:43]
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,32:43] * b_tile[0:11,0:43]
a_tile = A.get_tile(0,1)
b_tile = B.tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:21] * b_tile[11:32,0:43]
a_tile = A.get_tile(0,1)
b_tile = B.get_tile(2,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,21:43] * b_tile[0:22,0:43]
a_tile = A.get_tile(0,2)
b_tile = B.get_tile(2,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:10] * b_tile[22:32,0:43]
a_tile = A.get_tile(0,2)
b_tile = B.get_tile(3,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,10:42] * b_tile[0:32,0:43]
```

Universal RDMA Algorithm: Optimizations

Redundant Communications

- Instead of naively looping and overlapping tiles, **construct computation graph** of tiles to be multiplied.
- This graph can be traversed and modified to implement **computation/communication overlap**, prevent fetching tiles multiple times.

Rank 0

```
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,0:32] * B.tile(0,0)[0:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,32:43] * B.tile(1,0)[0:11,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,0:21] * B.tile(1,0)[11:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,21:43] * B.tile(2,0)[0:22,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,0:10] * B.tile(2,0)[22:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,10:42] * B.tile(3,0)[0:32,0:43]
```

```
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(0,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:32] * b_tile[0:32,0:43]
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,32:43] * b_tile[0:11,0:43]
a_tile = A.get_tile(0,1)
b_tile = B.tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:21] * b_tile[11:32,0:43]
a_tile = A.get_tile(0,1)
b_tile = B.get_tile(2,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,21:43] * b_tile[0:22,0:43]
a_tile = A.get_tile(0,2)
b_tile = B.get_tile(2,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:10] * b_tile[22:32,0:43]
a_tile = A.get_tile(0,2)
b_tile = B.get_tile(3,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,10:42] * b_tile[0:32,0:43]
```

Universal RDMA Algorithm: Optimizations

Redundant Communications

- Instead of naively looping and multiplying tiles, **construct computation graph** of tiles to be multiplied.
- This graph can be traversed and modified to implement **computation/communication overlap**, prevent fetching tiles multiple times

Rank 0

```
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,0:32] * B.tile(0,0)[0:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,32:43] * B.tile(1,0)[0:11,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,0:21] * B.tile(1,0)[11:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,21:43] * B.tile(2,0)[0:22,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,0:10] * B.tile(2,0)[22:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,10:42] * B.tile(3,0)[0:32,0:43]
```

```
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(0,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:32] * b_tile[0:32,0:43]
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,32:43] * b_tile[0:11,0:43]
a_tile = A.get_tile(0,1)
b_tile = B.tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:21] * b_tile[11:32,0:43]
a_tile = A.get_tile(0,1)
b_tile = B.get_tile(2,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,21:43] * b_tile[0:22,0:43]
a_tile = A.get_tile(0,2)
b_tile = B.get_tile(2,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:10] * b_tile[22:32,0:43]
a_tile = A.get_tile(0,2)
b_tile = B.get_tile(3,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,10:42] * b_tile[0:32,0:43]
```

Universal RDMA Algorithm: Optimizations

Redundant Communications

- Instead of naively fetching and multiplying tiles, **construct computation graph** of tiles to be multiplied.
- This graph can be used to modify to implement **computation/communication overlap**, prevent fetching tiles multiple times

Rank 0

```
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,0:32] * B.tile(0,0)[0:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,32:43] * B.tile(1,0)[0:11,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,0:21] * B.tile(1,0)[11:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,21:43] * B.tile(2,0)[0:22,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,0:10] * B.tile(2,0)[22:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,10:42] * B.tile(3,0)[0:32,0:43]
```

```
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(0,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:32] * b_tile[0:32,0:43]
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,32:43] * b_tile[0:11,0:43]
a_tile = A.get_tile(0,1)
b_tile = B.tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:21] * b_tile[11:32,0:43]
a_tile = A.get_tile(0,1)
b_tile = B.get_tile(2,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,21:43] * b_tile[0:22,0:43]
a_tile = A.get_tile(0,2)
b_tile = B.get_tile(2,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:10] * b_tile[22:32,0:43]
a_tile = A.get_tile(0,2)
b_tile = B.get_tile(3,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,10:42] * b_tile[0:32,0:43]
```

Universal RDMA Algorithm: Optimizations

Redundant Communications

- Instead of naively fetching and multiplying tiles, **construct computation graph** of tiles to be multiplied.
- This graph can be used to implement **computation/communication overlap**, prevent fetching tiles multiple times.

Rank 0

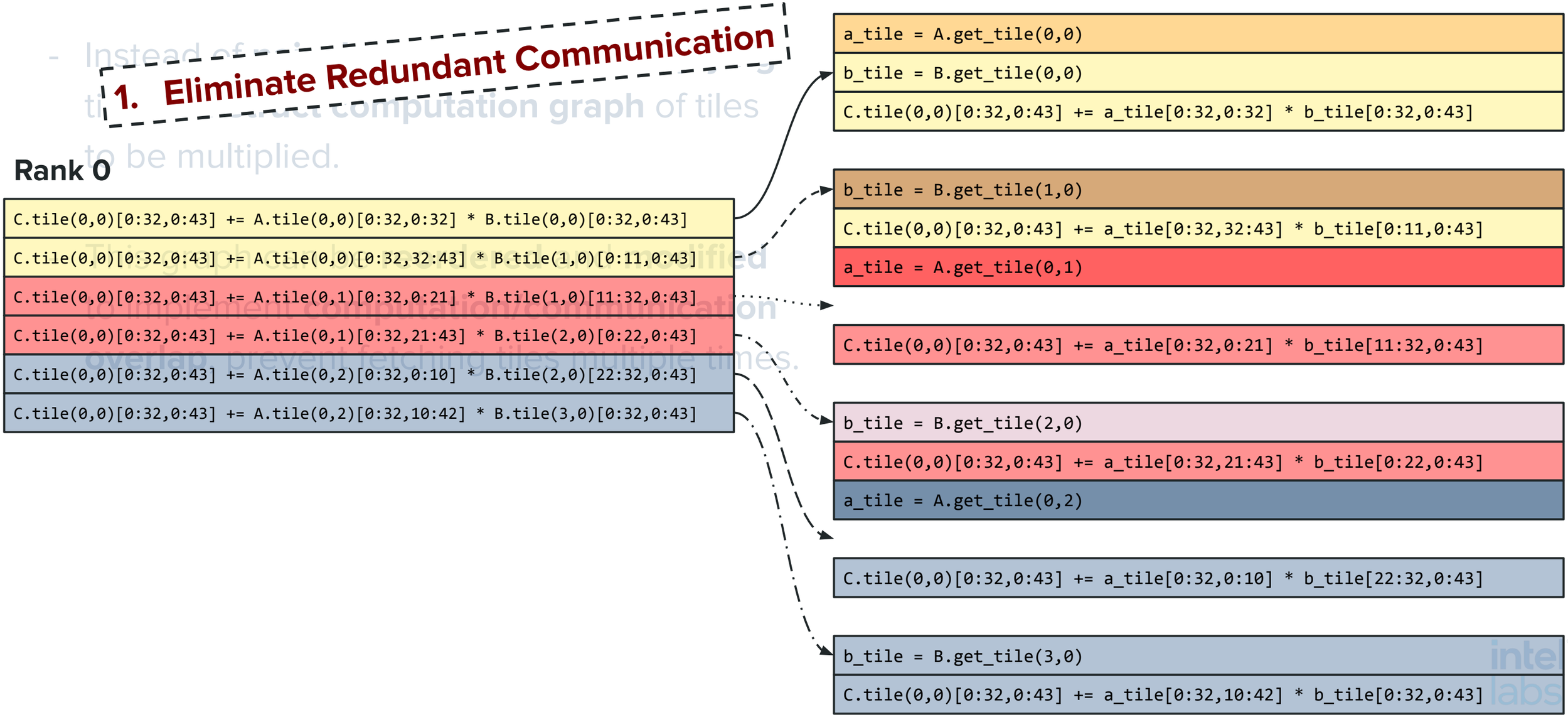
```
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,0:32] * B.tile(0,0)[0:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,32:43] * B.tile(1,0)[0:11,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,0:21] * B.tile(1,0)[11:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,21:43] * B.tile(2,0)[0:22,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,0:10] * B.tile(2,0)[22:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,10:42] * B.tile(3,0)[0:32,0:43]
```

```
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(0,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:32] * b_tile[0:32,0:43]
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,32:43] * b_tile[0:11,0:43]
a_tile = A.get_tile(0,1)
b_tile = B.tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:21] * b_tile[11:32,0:43]
a_tile = A.get_tile(0,1)
b_tile = B.get_tile(2,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,21:43] * b_tile[0:22,0:43]
a_tile = A.get_tile(0,2)
b_tile = B.get_tile(2,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:10] * b_tile[22:32,0:43]
a_tile = A.get_tile(0,2)
b_tile = B.get_tile(3,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,10:42] * b_tile[0:32,0:43]
```

Universal RDMA Algorithm: Optimizations

1. Eliminate Redundant Communication

- Instead of [illegible] the **exact computation graph** of tiles to be multiplied.
- This graph can be traversed and modified to implement computation/communication overlap, prevent fetching tiles multiple times.



Universal RDMA Algorithm: Optimizations

- Instead of **naively fetching** and **multiplying** tiles, **construct a computation graph** of tiles to be multiplied.
- This graph can be **traversed and modified** to implement **computation/communication overlap**, prevent fetching tiles multiple times

2. Prefetching

Rank 0

```
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,0:32] * B.tile(0,0)[0:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,0)[0:32,32:43] * B.tile(1,0)[0:11,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,0:21] * B.tile(1,0)[11:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,1)[0:32,21:43] * B.tile(2,0)[0:22,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,0:10] * B.tile(2,0)[22:32,0:43]
C.tile(0,0)[0:32,0:43] += A.tile(0,2)[0:32,10:42] * B.tile(3,0)[0:32,0:43]
```

```
a_tile = A.get_tile(0,0)
b_tile = B.get_tile(0,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:32] * b_tile[0:32,0:43]
```

```
b_tile = B.get_tile(1,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,32:43] * b_tile[0:11,0:43]
a_tile = A.get_tile(0,1)
```

```
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:21] * b_tile[11:32,0:43]
```

```
b_tile = B.get_tile(2,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,21:43] * b_tile[0:22,0:43]
a_tile = A.get_tile(0,2)
```

```
C.tile(0,0)[0:32,0:43] += a_tile[0:32,0:10] * b_tile[22:32,0:43]
```

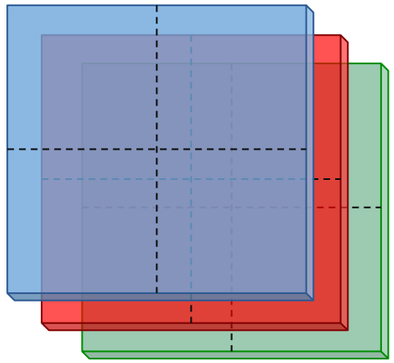
```
b_tile = B.get_tile(3,0)
C.tile(0,0)[0:32,0:43] += a_tile[0:32,10:42] * b_tile[0:32,0:43]
```

Universal RDMA Algorithm: Optimizations

- We can also support **replication** of one or more matrices.

Replication Factor c = 3
(3 Copies)

Each replica split
between 4 GPUs.



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Universal RDMA Algorithm: Optimizations

- We can also support **replication** of one or more matrices.

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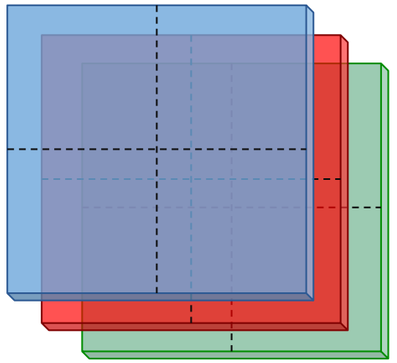
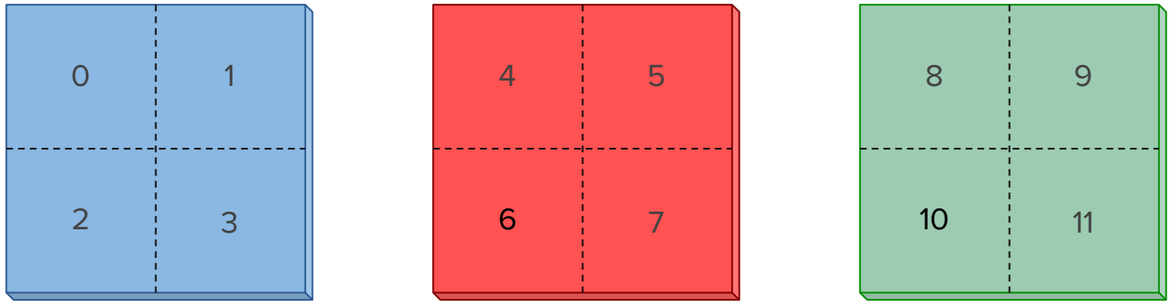
(3 Copies)



Each replica split
between 4 GPUs.



Universal RDMA Algorithm: Optimizations

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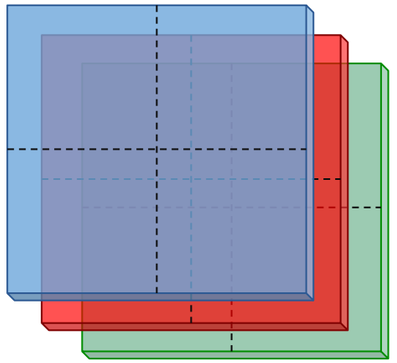

Each replica split
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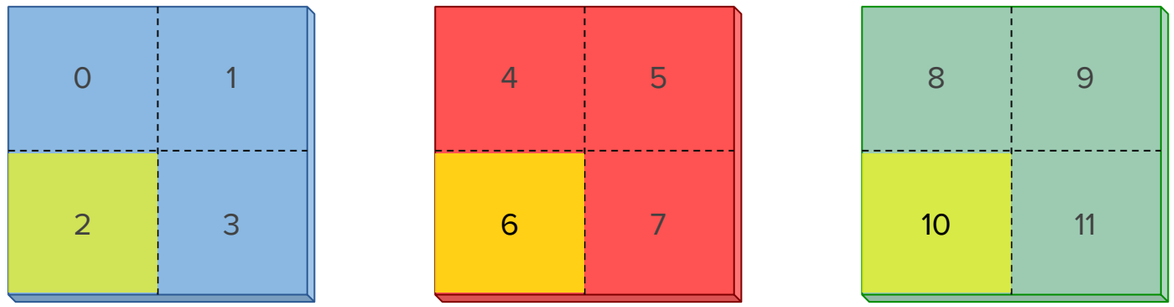

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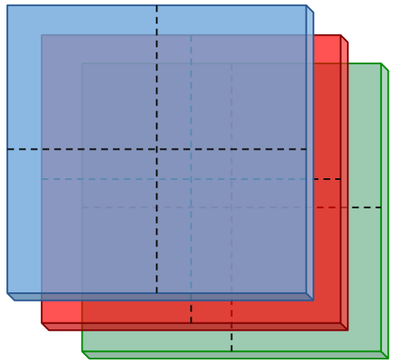
Universal RDMA Algorithm: Optimizations

- We can also support **replication** of one or more matrices.

- This requires **decomposing** the inner loop (split-k style).

Replication Factor c = 3
(3 Copies)



Each replica split
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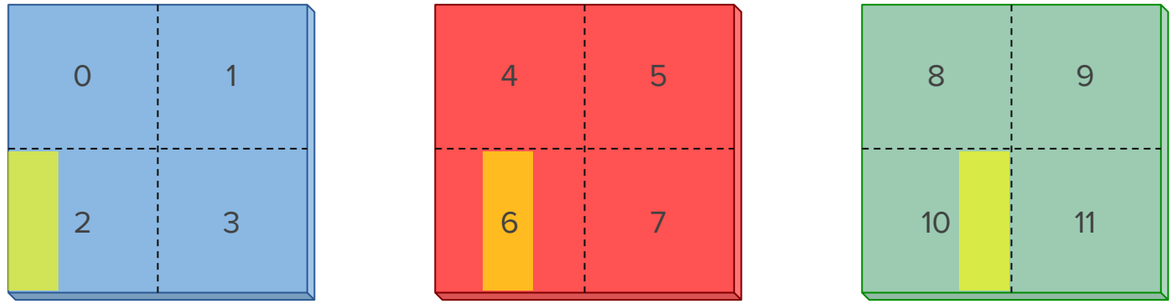

Universal RDMA Algorithm: Optimizations

- We can also support **replication** of one or more matrices.

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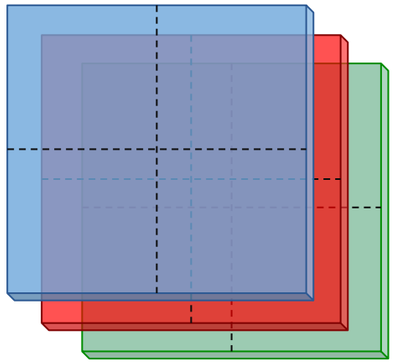
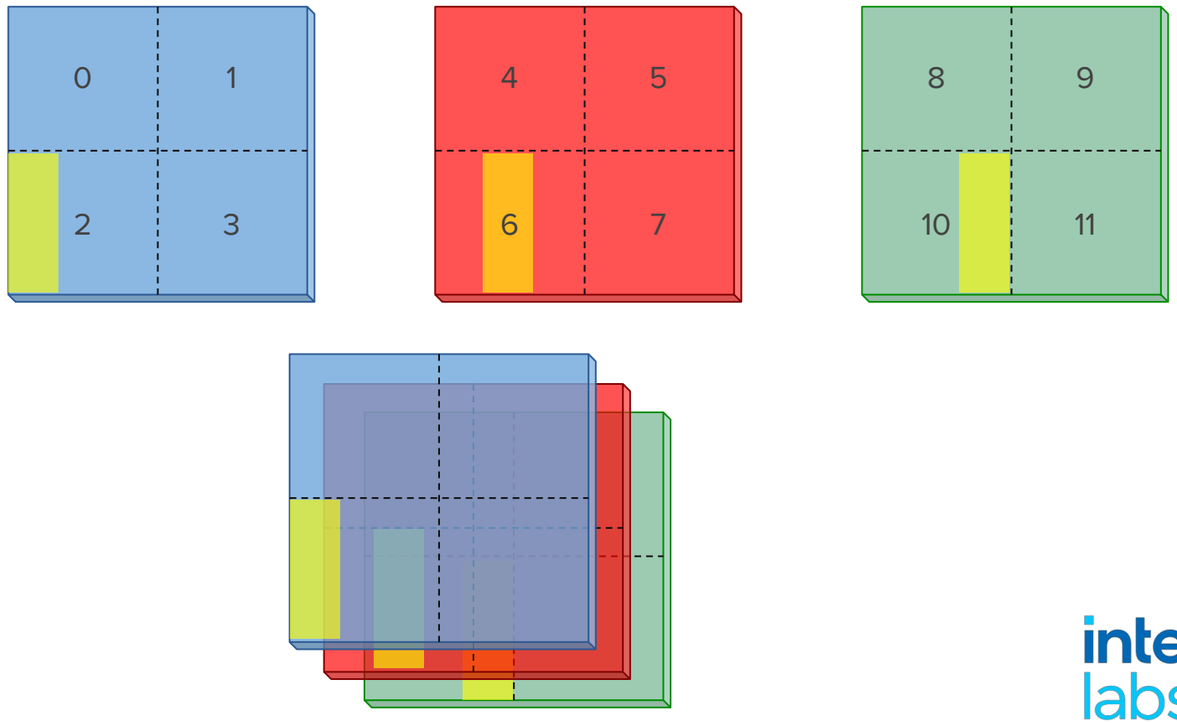
Replication Factor c = 3
(3 Copies)



Each replica split
between 4 GPUs.



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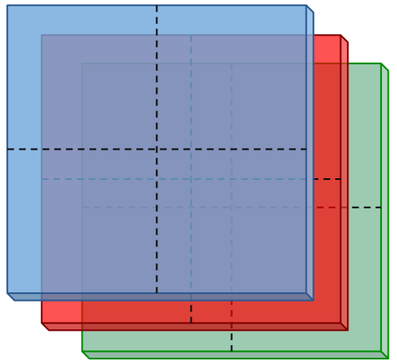
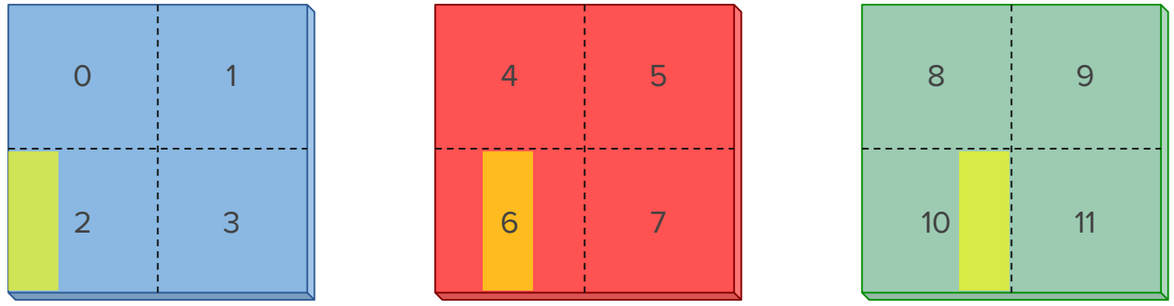
Universal RDMA Algorithm: Optimizations

- We can also support **replication** of one or more matrices.
- This requires decomposing the inner loop (split-k style).
- Followed by an **AllReduce** to accumulate the final result.

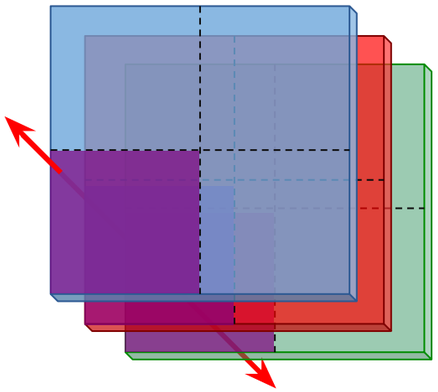
Replication Factor c = 3
(3 Copies)



Each replica split
between 4 GPUs.



AllReduce!



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Universal RDMA Algorithm: Replication

- Generating IR for replicated matrix multiplication is **trivial** under this model.

- Simply **split the iteration space** based on the **replication factor**.

Pseudocode: Produce IR

```
# Each process produces a list of operations to perform.
for idx,block in 'blocks of C I own':
    # Compute the bounds of my tile
    c_bounds = c.tile_bounds(idx)

    # Find tiles of A that overlap.
    a_tiles = a.overlapping_tiles(c_bounds[0],:)

    for a_idx in a_tiles:
        # Find tiles of B that overlap with our output bounds (of C)
        # *And* the current A tile's bounds.
        a_tile_bounds = a.tile_bounds(a_idx)
        b_tiles = b.overlapping_tiles(a_tile_bounds[1], c_bounds[1])

        for b_idx in b_tiles:
            # Emit instruction that multiplies tiles of
            # A, B, and C, including the appropriate slicing
            # of these tiles.
```

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Universal RDMA Algorithm: Replication

- Generating IR for replicated matrix multiplication is **trivial** under this model.

- Simply **split the iteration space** based on the **replication factor**.

Pseudocode: Produce IR

```
# Each process produces a list of operations to perform.
for idx,block in 'blocks of C I own':
    # Compute the bounds of my tile
    c_bounds = c.tile_bounds(idx)

    # Find tiles of A that overlap.
    my_split_k = slice(my_replica_index*slice_size,
                       (my_replica_index+1)*slice_size)
    a_tiles = a.overlapping_tiles(c_bounds[0],my_split_k)

    for a_idx in a_tiles:
        # Find tiles of B that overlap with our output bounds (of C)
        # *And* the current A tile's bounds.
        a_tile_bounds = a.tile_bounds(a_idx)
        b_tiles = b.overlapping_tiles(a_tile_bounds[1], c_bounds[1])

        for b_idx in b_tiles:
            # Emit instruction that multiplies tiles of
            # A, B, and C, including the appropriate slicing
            # of these tiles.
```

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Universal RDMA Algorithm: Replication

- Generating IR for replicated matrix multiplication is **trivial** under this model.
- Simply **split the iteration space** based on the **replication factor**.
- If **A and B** are replicated, **grab from local replica** (otherwise, just retrieve).

Pseudocode: Produce IR

```
# Each process produces a list of operations to perform.
for idx,block in 'blocks of C I own':
    # Compute the bounds of my tile
    c_bounds = c.tile_bounds(idx)

    # Find tiles of A that overlap.
    my_split_k = slice(my_replica_index*slice_size,
                       (my_replica_index+1)*slice_size)
    a_tiles = a.overlapping_tiles(c_bounds[0],my_split_k)

    for a_idx in a_tiles:
        # Find tiles of B that overlap with our output bounds (of C)
        # *And* the current A tile's bounds.
        a_tile_bounds = a.tile_bounds(a_idx)
        b_tiles = b.overlapping_tiles(a_tile_bounds[1], c_bounds[1])

        for b_idx in b_tiles:
            # Emit instruction that multiplies tiles of
            # A, B, and C, including the appropriate slicing
            # of these tiles.
```

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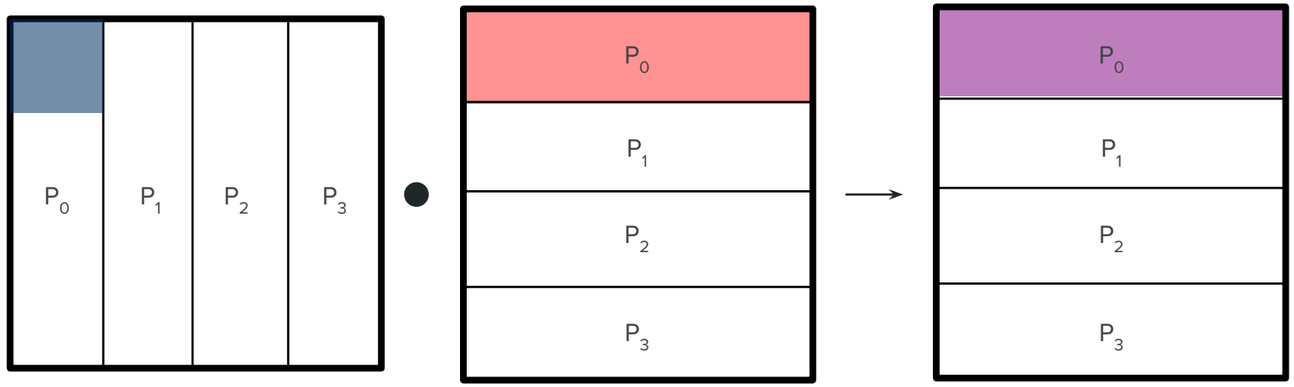
Remote Updates: Stationary Methods

There are **three ways** to schedule work:

- Stationary **A**
- Stationary **B**
- Stationary **C**



Stationary A and B require **remote accumulations**

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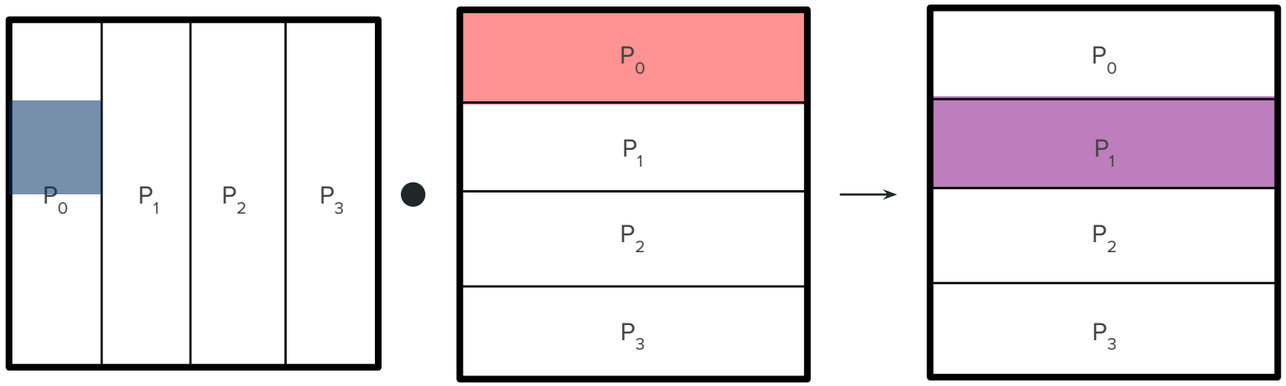

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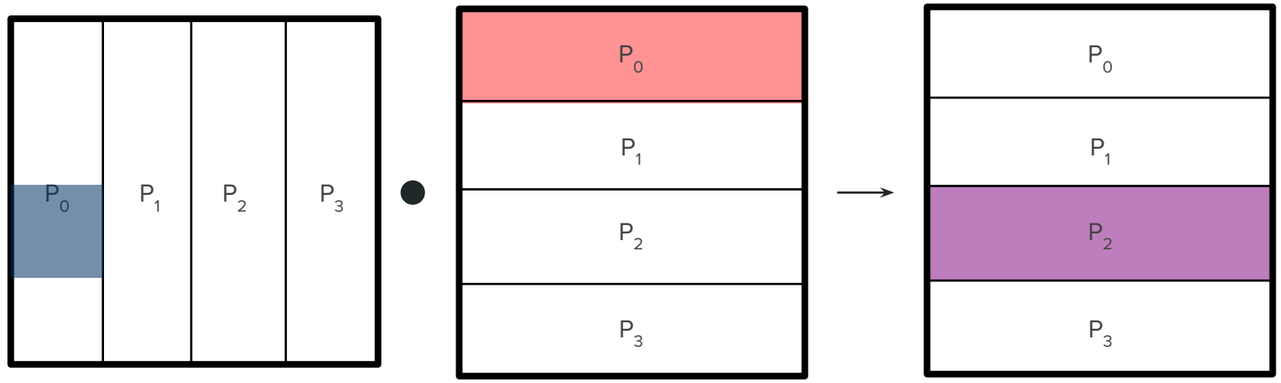

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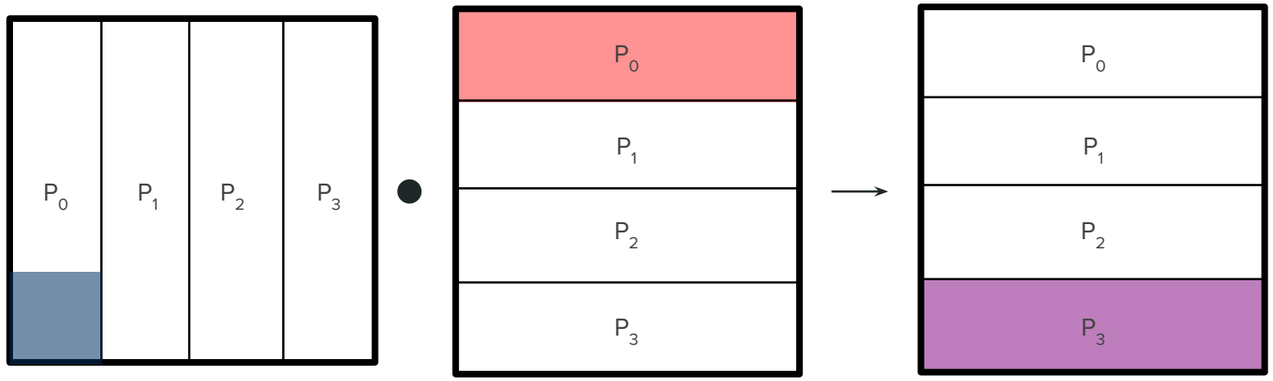

Stationary A and B require **remote accumulations**

Remote Updates: Accumulation Kernels

- With memory semantic **intra-node networks** (Xe Link, NVLink, Infinity Fabric, etc.), we can perform remote updates with a kernel

- These are **relatively efficient**, even with fetch-and-add atomics (> 80% of peak BW)

```
/* . . . */

void accumulate_tile(index tile_index, matrix_view<T> view) {

    T* remote_data = tile(tile_index).data();
    T* local_data = view.data();

    return parallel_for(
        queue(), view.shape()[0] * ld,
        [=](auto idx) {
          auto column_index = idx % ld;
          if (column_index < view.shape()[1]) {
            // atomic fetch-and-add
            remote_data[idx] += local_data[idx];
          }
        });
}
```

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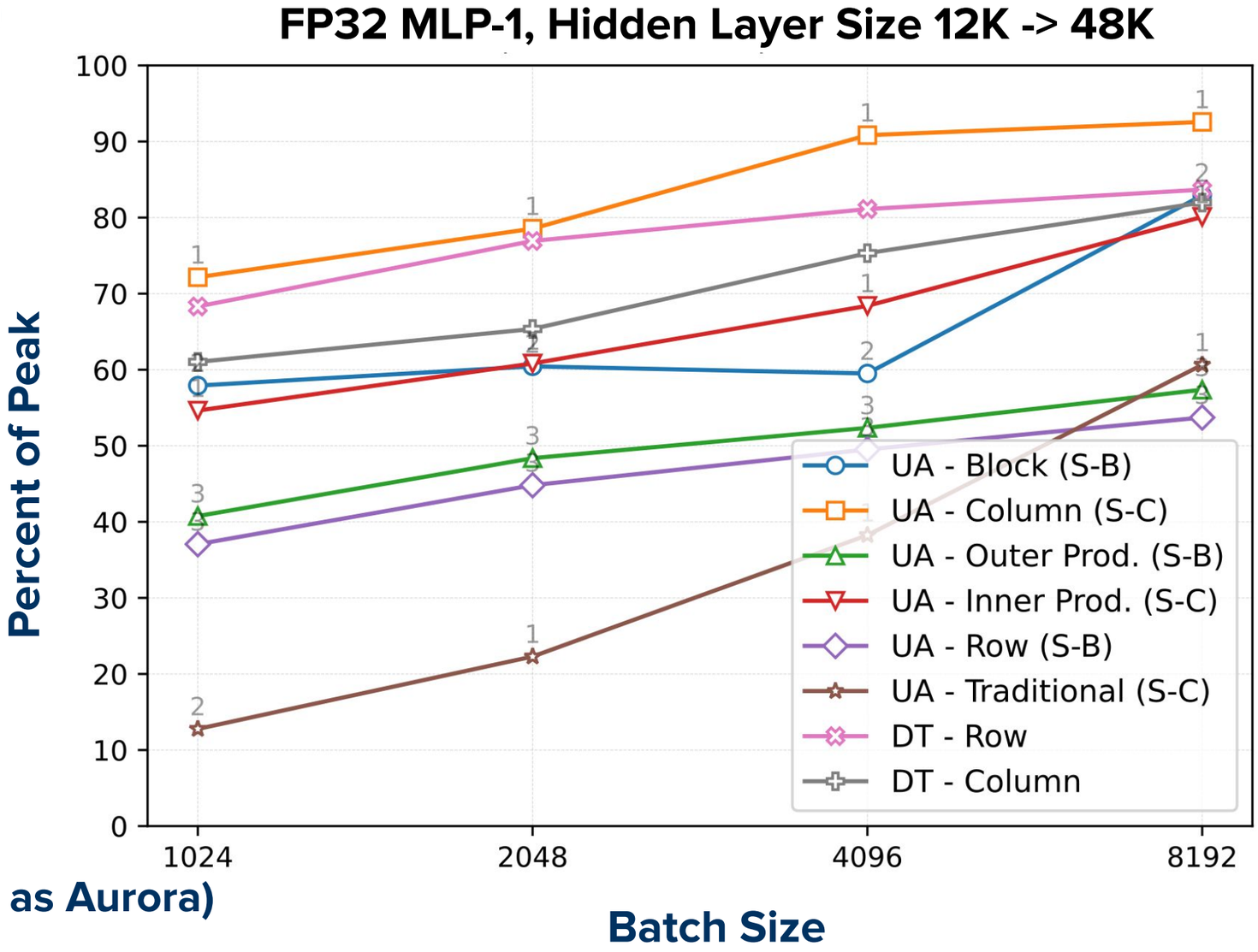
Transformer Layer MLP (1)

- We can set up problems with a **variety of distributions** and feed all of them into the universal algorithm.

- Different problem sizes favor different distributions.



Run on Borealis **(Same Configuration as Aurora)**

12 PVC Tiles per Node

20 GB/s Xe Link Between Tiles

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Transformer Layer MLP (2)

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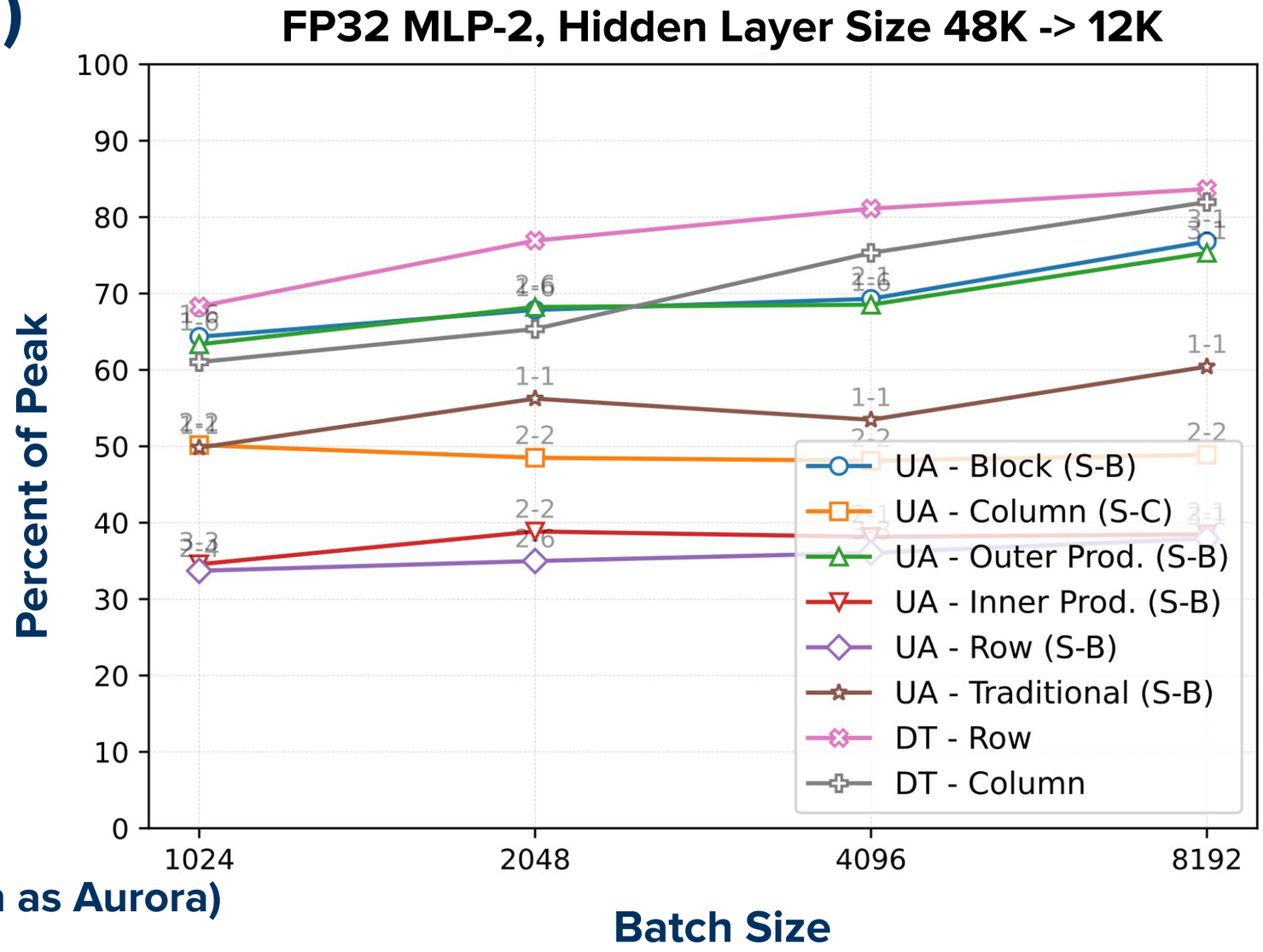

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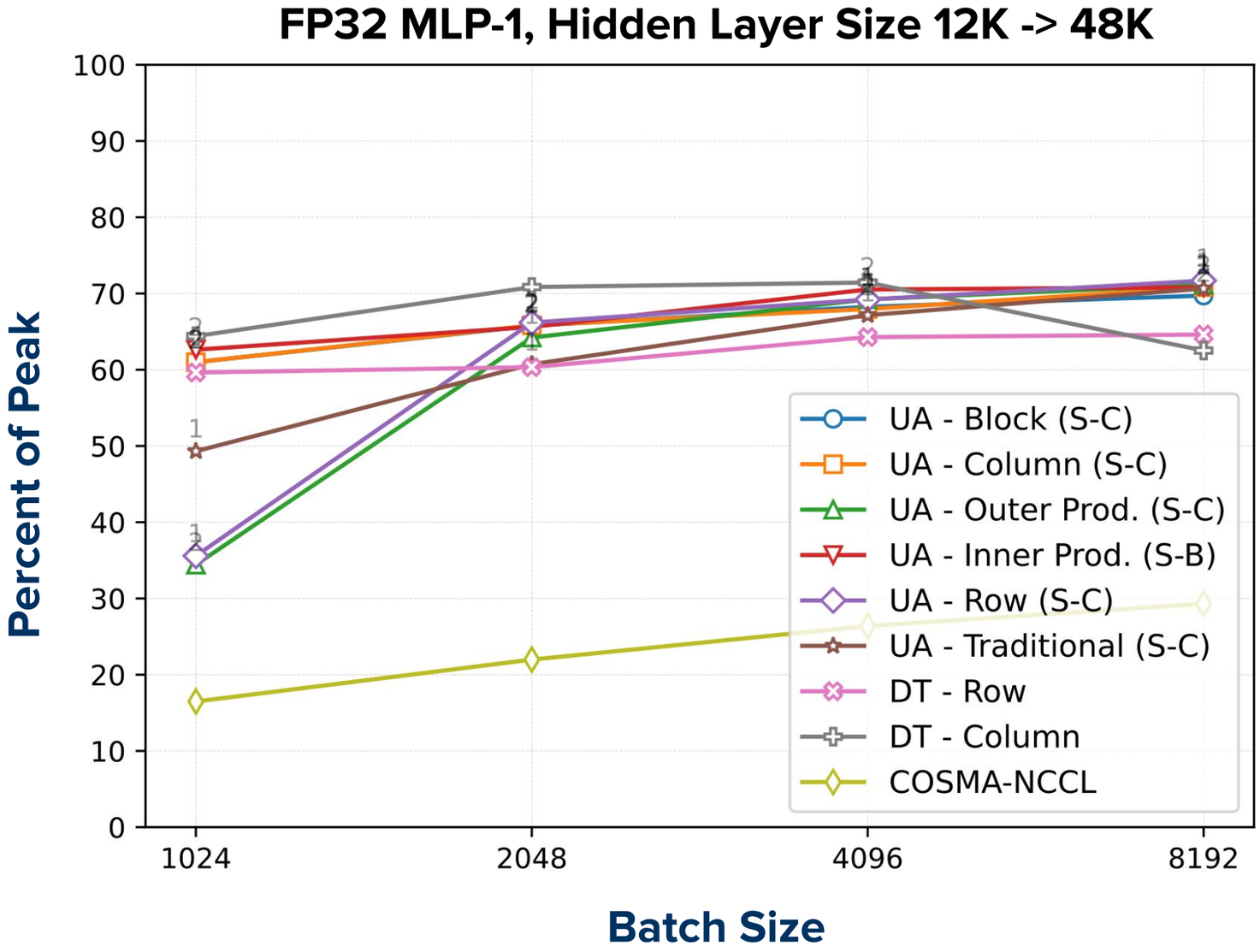

Run on H100 System

8 H100 GPUs per Node

450 GB/s NVLink Between Tiles

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Transformer Layer MLP (2)

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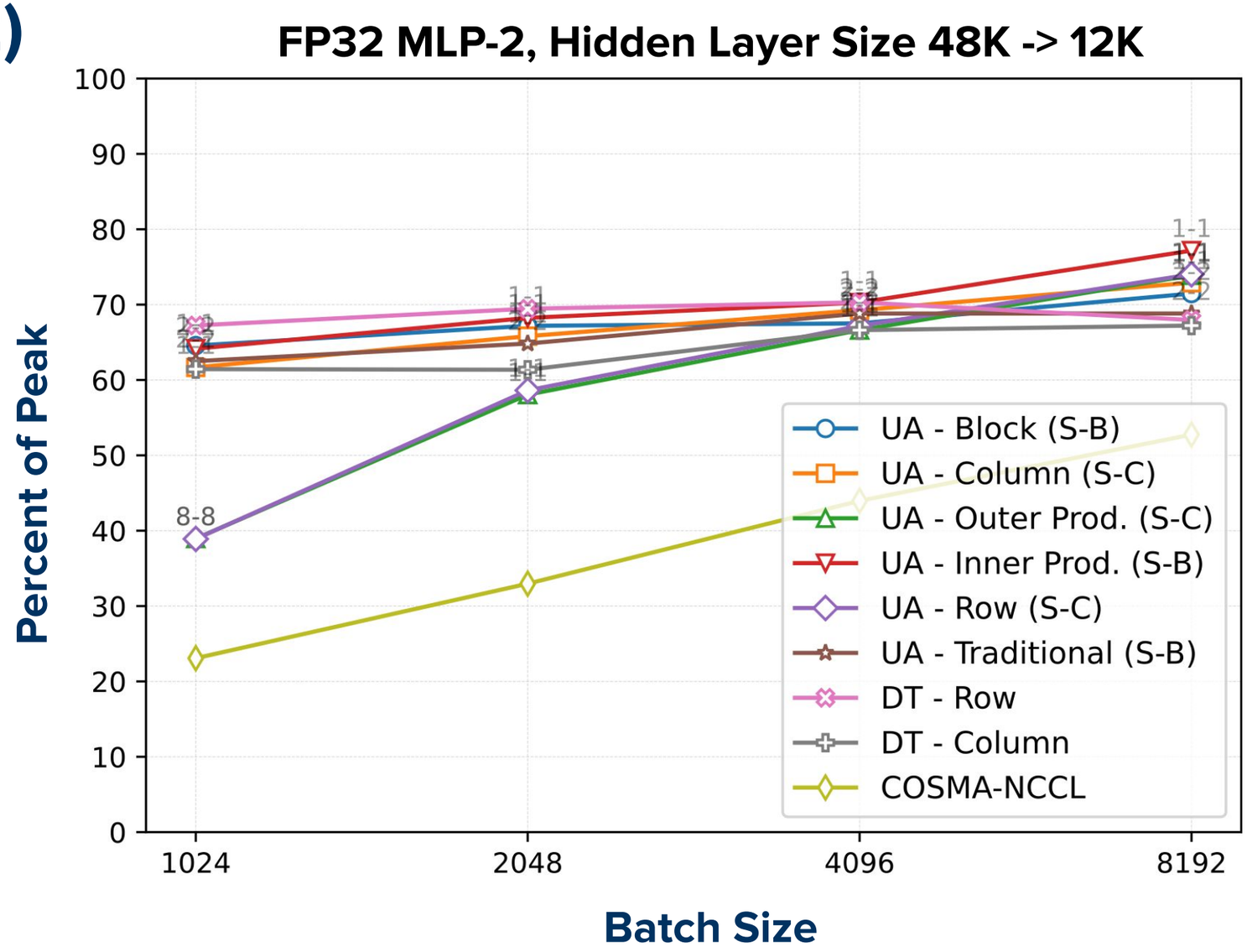

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Future Work

Unanswered question: **which partitioning should I use**?

Select partitioning to minimize both **communication cost** *and* **local GEMM cost**.

The final frontier: **GPU-initiated communication**.

Conclusions

Exploring the **full space of partitioning and replication options** is important for achieving high performance with Matmul.

We can use **one-sided communication** to support all possible partitionings, achieving good performance.